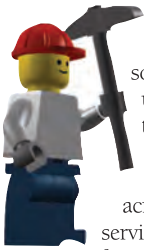
Google Squared
Google's experiments with semantic search are at www.google.com/squared

AJAX
Read up on why the term was coined, and what it means at www.thinkdigit.com/d/29349/

software should have taken long ago. So, what's next?

The web has a few shortcomings, which will get amplified rapidly now. Searching for Web 2.0 sites is already a difficult thing. A search engine is made to crawl hyperlinked pages, and throw up the answer. Almost any Google search now gives you a Wikipedia page as the first or second link. Because of the way Web 2.0 has evolved, we are, or soon will be, back to where we started, in the pre-Google days, without some kind of a search engine to make sense of the web. Spiders are not designed to crawl content-rich sites, go to the deep back end of web-based applications, because the web is no longer made up of mere hyperlinks. Consolidating a database of web services is the next big challenge for search engines. There is only so far that the tags and metadata of today can take you. Internet search will have to undergo a radical change to keep pace with Web 2.0. In fact, what it means to search the web itself, might change radically.

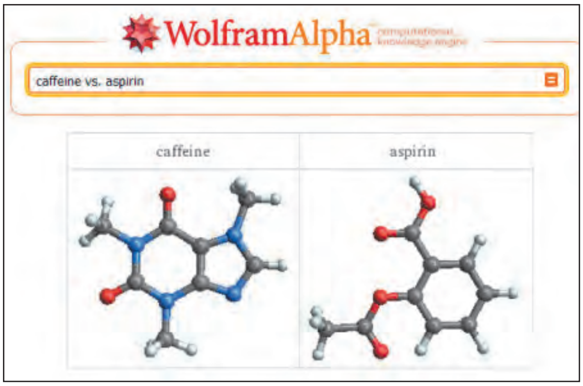


Plans for the future

The next generation of social networking will probably use the current services as the back-end. The "front" will have to consolidate the web activities of users across a wide range of web services. Say a common service for updating your status on both Twitter and Facebook is one of the most basic functionality to start off with, something that a service called Plurk already offers. The next step is auto-updating the status message based on the current activity of the user. To a limited extent, there are applications that say monitor your activity on Last.fm and re-submit the scrobbles as a status update on Twitter. A new generation of customisable middlemen, engines that let you take feeds, remix them, and feed them back to another service, or the same service are in the offing. Again, bots on Twitter do this to a limited extent. Swearbot, for example, monitors all instances of swearing on the Tweetosphere, and reprimands the swearer for using strong language. Flock is a web browser that lets you plug in to Orkut, Flickr, Facebook, YouTube, Twitter and other Web 2.0 services from one easy interface. Much more of that is headed your way. Theorists are talking about a "lifestream", a steady flow of data of what you are doing online, used by a multitude of services, for a multitude of applications, but the lifestream itself being invisible as a whole.

Another important line that will be breached is the difference between online and offline applications. Web 3.0 applications will fuse seamlessly with desktop applications in a manner that there is little difference between the two. Your data will be accesible from any computer, and you can drag-drop content from the web to your system folders and items from your desktop to say, an attachment in your email. Google Wave, will offer some of these functions when it is released to the common public. Another important line that will get blurred, but over a period of time, is the difference between using your computer and your mobile phone. The interfaces are pretty low-key even on high end mobile phones with customised applications for Web 2.0 services, but these will get better. You can expect to be connected and online all the time, and use the Web 2.0 in a more personalised way. A large role will be played by the search engines of Web 3.0.

Search engines will have to become much more smarter, presenting portals based on the search term as their result, instead of a set of pages. For example, if you search for "India", the search engine should throw up a page which has snippets from wikis with a different focus (say travel, encyclopedic, and pop culture), feeds from recent blog entries, tweets and plurks, as well as photos from different pools. Possibly, airline tickets to popular airports, festivals in the country, and recent news from the location could also show up. The specifics of this would depend on user preference and history. Additionaly, the content will have to change depending on whether the search is made from within India our outside. There are many more demands that users will have from search engines of Web 3.0. If you walk into a store, click a photograph of something on sale, and search using that photograph, the search engine should be able to throw up prices on nearby stores, and directions to the cheapest location of sale. The back end should be pretty robust, so as not to disappoint out of the way requests by users who really need it. Say, a person spots a great looking city while flying, and photographs it from the air, the search engine should be good enough to resolve and identify the location using a pretty low-res mobile phone camera - without any appreciable load time at that. No search engine does all this yet. Wolfram-Alpha is a step in this direction. The functionality of the service is very limited, and falls drastically short of its claims, but that just goes to show the staggering task in designing the next generation of search engines. A single search engine for all these applications would be great, and no guesses for which one will try the hardest to get there, but smaller engines with highly specialised and robust application can get ahead. For example, both iTunes and TrackID on Sony Ericsson phones use Gracenote's database and search facility to identify songs. Gracenote is well ahead of Google when it comes to searching using snippets of songs, or their tags.



Wolfram Alpha's search results are not always what most people expect

Web 3.0 will require some kind of system where all the applications can pool their resources. Some common language such as XML, that will describe the content generated by the users, and their relation with each other. There are no standards as yet for this language, but when it comes, it will help the web understand us better.

There are two divergent paths that the web is taking. One approach is to consolidate everything in one central location, and the other is to distribute everything to the user. Google Docs, On Live and AMD's Fusion render cloud are examples where data is centrally located, and worked on by remote applications. This includes applications where a game for any console can be played on any console. Another approach, drastically different, is Opera's Unite browser, and software such as Freenet. Both of these bring the power of web hosting to individual computers. Applications such as Seti@Home have long been using distributed computing for processing large amounts of data. Web 3.0 will incorporate both of these approaches, further blurring the lines between personal and public digital space.

There is a flip side to all of this as well. These trends are not exciting to everyone. There is a small minority of people who don't really approve of the direction in which the web is going at all. According to these people, blogging just introduced a whole bunch of clutter, Twitter is an irritating phenomenon, and all of these are passing fads. Privacy, and the use of personal data for advertising is another big concern, even if this is done by machines without human interaction. These concerns can not stop the web from going where it will though, and for the digital community, the world continues to grow smaller by the second.

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A more detailed look at Wolfram|Alfa
http://www.thinkdigit.com/d/65572/